Online Ammonia Sensor User Manual

Xi’an Desun Uniwill Electronic Technology Co., Ltd.

Please Read This Manual Carefully Before Use.

Preface

Thank you very much for choosing our company!

Please read this manual carefully before using this product, and save it for reference. Please observe the operating procedures and precautions in this manual.

To ensure that the after-sale protection provided by this instrument is effective, please do not use and maintain the instrument in any way other than specified in this manual.

As the failure to observe the precautions specified in this operation manual, any failures and losses caused are not covered by the manufacturer's warranty, and the manufacturer does not assume any related responsibility. Please keep all documents in a safe place. If in doubt, please contact our after-sales service department.

When receiving the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to shipping. If damage is found, please contact Desun Uniwill after-sales service department and keep the packaging for return.

When the instrument fails, please do not repair it by yourself, please contact our after- sales service department.

Online NH4-N Ammonia Nitrogen Sensor

I. Overview

The Ammonia nitrogen sensor is an online device for detecting ammonia nitrogen (NH4+-N) levels in water and is suitable for use in a variety of water bodies, including lakes, streams, groundwater and waste water.

The sensor measures ammonia nitrogen values by the electrode method. The NH4+ electrode provides the main measurement, which measures ammonia ions (NH4+) with automatic temperature and pH compensation and an environmentally friendly design.

The sensor is waterproof to IP68 and supports the MODBUS protocol. Higher accuracy, wider measuring range and greater stability, It Can be directly installed, compared with traditional ammonia nitrogen analyzer, more economical and environmentally friendly, convenient and fast. It adopts RS485 output and supports Modbus for easy integration.

Features

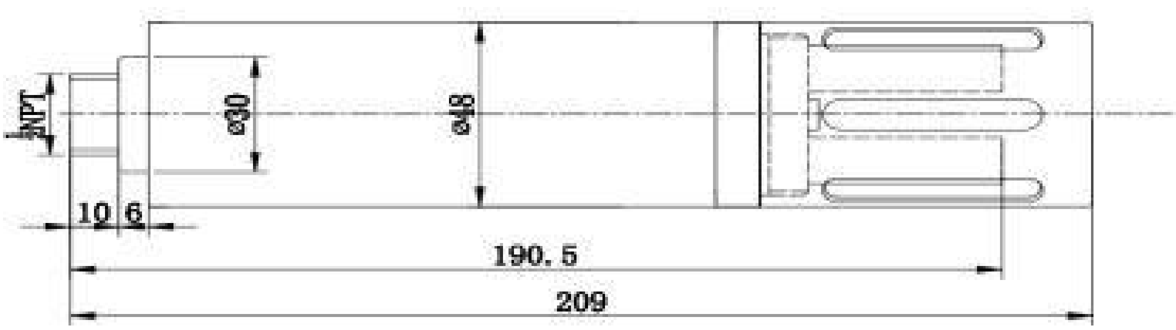
- ✧ With pH, potassium ion and temperature compensation, suitable for monitoring a wide range of water quality environments.
- ✧ Each electrode can be replaced independently, easy to operate and low maintenance.
- ✧ Accurate measurements, low detection limits and low long-term drift.
- ✧ Fast response time, reflecting changes in the water sample in a timely manner.
- ✧ NPT temperature compensation, accuracy up to ±0.1°C.
- ✧ Corrosion-resistant housing for long-term underwater operation and compact construction.
- ✧ RS485 communication interface, standard Modbus protocol, easy to integrate.

Parameters

Ammonia nitrogen range	0-1000mg/L NH4-N
Ammonia nitrogen precision	3%
Ammonia nitrogen resolution	0.1 mg/L
pH range	4-10
pH accuracy	±0.1
pH resolution	0.01
Protection level	IP68
Deepest depth	10 meters underwater

Temperature range	0 ~ 50 °C
Output signal	Support RS-485, MODBUS protocol
Assembly	Input type
Power supply	DC6~12VDC±10%, current <50mA
Cable length	5 meters (default), can be customized
Housing material	POM

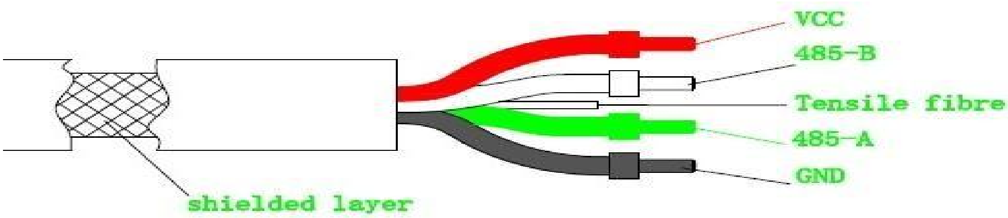
II. Dimensions



III. Cable Information for Sensors

4 wire AWG-24 or AWG-26 shielding wire. OD=5mm

- Red line----power supply (VCC)
- White line---- 485 data -B (485-B)
- Yellow line---- 485 data -A (485-A)
- Black line----ground (GND)
- Bare line---- shielding layer



The power supply must be DC 6-12V +/-5%, current <50mA.

IV. Maintenance Methods and Common Problems

Maintenance Schedule and Method

Maintenance Schedule

It is recommended to periodically calibrate or clean the sensor surface according to the actual application scenario.

Maintenance Tasks	Recommended Maintenance Frequency
Calibrate the sensor (if required by the competent authority)	According to the maintenance schedule required by the competent authority

Maintenance Method

Sensor Maintenance

1) **External surface of the sensor:** Wash the external surface of the sensor with tap water. If debris remains, wipe it with a moist soft cloth.
2) **Check the cable of the sensor:** The cable should not be tight during normal work, otherwise the internal wires of the cable will be easily broken and the sensor will not work properly.
3) **Sensor storage:** When the sensor is not in use, you should fasten the black plastic cap and check whether the sponge inside is wet. If it is not wet enough, please add 3 mol/L potassium chloride solution and let the electrode be stored in the solution with potassium chloride.

Precautions

The probe contains sensitive optical and electronic components. Ensure that the probe is not subject to severe mechanical impact. Damage to the sensor due to impact or manpower will not be covered by the warranty. Water ingress to the sensor due to replacement of the pH electrode is not covered by the warranty.

Common Problems

Error	Possible causes	Solutions
The operation interface cannot be connected or the measurement results are not displayed	Controller and cable connection error	Reconnect the controller and cables
	Cable failure	Please contact us
Measured values are too high, too low, or the values remain unstable	Measurement shifted	**Recalibration**
	Water has entered the product	Please contact us
	The pH electrolyte is exhausted	Replace the pH electrochemical part

V. Description of Details




This sensor has four electrode heads and they are all different in function.

1. Ammonium ion
2. pH/ORP
3. Reference electrode
4. Optional ion: e.g. potassium ion.

VI. Sensor Installation

Before installation, please check the appearance of the cable for damage and disperse the cable.

1. Input sensors should be used to avoid the sensor due to water flow caused by the sensor hit the wall or other facilities, if the water flow rate is too fast need to fix the sensor.
2. There should not be any obstructions in front of the measurement, such as fluorometric sensors should avoid light exposure.
3. The installation of the sensor should not exceed 2 meters from the horizontal water depth position, and according to the fluctuations of the water flow, the installation position should not be less than 30 cm water depth. (According to the measured water body can be adjusted)
4. The sensor should be installed in the location without air bubbles, to prevent the impact of air bubbles on the sensor measurement data.
5. The sensor should not be used to lift the sensor itself cable, install cable protection devices to prevent the cable and the sensor between the taut tension is too large, resulting in cable breakage.
6. The sensor should not be installed with the measuring surface facing upward, can be installed vertically downward or horizontally.
7. If the sensor is used in the water diversion tank, please consult the company's sales

staff to confirm the specific requirements of installation.

VII. Quality Assurance

Our company assures its first-hand buyers that there will not be any product defects caused by substandard materials or factory manufacturing in the 3 months after shipment. If a defect is found within the 3-month warranty period, Desun Uniwilli Electronics promises to repair or replace the defective product or return the payment in addition to the first shipping and related formalities. Any product repaired or replaced during the warranty period will only enjoy the remaining warranty period of the original product.

This warranty does not apply to consumables such as consumable parts (including but not limited to lamps, pipes, etc.).

Contact Desun Uniwill Electronics or your agent to begin technical support during the warranty period.

After receiving the customer's questions about product quality, our company will confirm whether the product needs to be repaired within two weeks; products that have not been approved for repair cannot be returned.

VIII. Warranty Limit

This warranty does not cover the following:

- Damage caused by force majeure, natural disasters, social unrest, war (published or unpublished), terrorism, civil war, or any government coercion
- Damage caused by improper use, negligence, accident or improper application and installation.
- Freight charges for shipping the goods back to Desun Uniwill
- Expedited or express shipping for parts or products covered by the warranty
- Travel expenses for local warranty repairs

This warranty covers all aspects of our company's warranty regarding its products. This warranty constitutes the final, complete and exclusive statement of the terms of the warranty, and no one or agent is authorized to formulate other warranties in the name of Disen Hezhi Electronics.

The remedies such as repair, replacement, or refund of the payment as described above are special cases that do not violate this quality guarantee. The remedies such as replacement or refund of the payment are targeted at our products. Based on strict liability or other legal theories, Desun Uniwill Electronics is not responsible for any other damage caused by product defects or negligent operation, including the subsequent damage caused by causality with these situations.